

Vishal Maurya

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EDUCATION

Indian Institute of Technology Kharagpur

Integrated B.Sc. & M.Sc. in Physics (8.02/10)

Advisor: Prof. Kannabiran Seshasayanan

Thesis title: Comparison between Bousinesq and Anelastic approximation while modelling MHD equations

West Bengal, India

Jul'19-May'24

WORK EXPERIENCE

Lightfury Games

May'24-Present

Engineer-2 (Founding team) at [ECricket](#)

- Engineered **custom physics engine**, optimized for multiplayer experience and runtime physics behaviour switching
- Designed an algorithm for **deterministic collision responses** between animation driven bat and physics driven ball
- Constructed user based batting/bowling **AI algorithms**, using **state trees** resulting **30%** increase in user engagement
- Developed an infrastructure for **scalable multiplayer experience** ranging from 1vsAI to 11vs11 (XvsY) for the game
- Built a **simulation** setup for recording batting/bowling/meta inputs, collision responses and improve the algorithm

AuraML

Jan'24-Apr'24

Software Development Intern at [AuraML](#)

- Utilized **Pixel streaming**, giving users access to customize their environment and generate images for their ML models
- Debugged issues related to memory and performance, using integrated debuggers in Unreal Engine and VS community
- Designed procedural generated warehouses using **PCG graphs** in Unreal engine 5, to generate synthetic datasets
- Programmed realistic object spawning logic in **blueprints**, such that user can create images as close to real environment

Mayhem Studios

Jun'23-Aug'23

Unreal Developer Intern at [Underworld Gang Wars](#)

- Engineered customizable in-game features accessible like One-tap ADS, quick scope, peek & scope accessible via settings
- Crafted a dynamic free fall feature, complete with VFX, SFX and fall damage mechanics, intensifying player experience
- Integrated engaging Sound effects for footsteps, swimming, throwables, consumables actions and environmental sounds
- Drove the integration of **Quilkin**, a non-transparent UDP proxy, improving network security and capturing telemetry

Supergaming

May'22-July'22

Game Developer Intern at [Tower Conquest: Metaverse](#)

- Designed UI screens for PvE and PvP game modes, and description tool-tips for UI elements in **Unity Engine**
- Implemented gameplay features using **Finite State Machines** in **Web3** based game **Tower Conquest: Metaverse**
- Resolved bugs related to **synchronization** and parameters governing toughness using Visual Studio's **debugger**
- Integrated with back-end server APIs and utilized JSON files to populate user and game assets-related information

PROJECTS

Balance | Mini Jam 84: Bubbles

Objective: To create a game based on Bubbles theme in the given time

- Created a timer-based game, where players have to tap on bubbles of gases to maintain an optimal level of each gas
- Delivered playable **WebGL** build of the game in 72 hours and was ranked **69th** among **600+** participants worldwide
- Playable WebGL build was published on [itch.io](#) and can be played using link <https://mvishal2002.itch.io/balance>

TECHNICAL SKILLS

- Programming:** C++, C, C, Python, JavaScript, Lua, HTML
- Game Engines & Frameworks:** Unreal Engine 5, Unity, LÖVE 2D, React
- Development & Version Control:** Perforce, Git, Github, TortoiseSVN, Sourcetree
- Development Tools:** JetBrains Rider, Visual Studio Code
- Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Json.NET

POSITIONS OF RESPONSIBILITY

Software Team Member, Computer Graphics Lab | IIT Kharagpur

Nov'19-Apr'21

Part of 20-member team involved in making Computer Graphics Applications and Games

- Contributed in gameplay features of the game **Ricksy Run** built on Unity engine having **35K+** downloads on Playstore
- Organized game development workshop to teach basics of Unity to **200+** students in collaboration with Kshitij 2020